

Homework - I

Statistics III : Multivariate Data and Regression

Utkarsh Bharadwaj

Roll Number: BSD-BG-2521

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Question 1

Suppose the random variable Y comes from a true model

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

where ϵ has mean zero and variance σ^2 . Suppose we have independent observations (y_i, x_{1i}, x_{2i}) , $i = 1, \dots, n$ where x_{1i}, x_{2i} are fixed. We fit a linear model of y on x_1 (only) by least squares to obtain the estimators $\hat{\alpha}$ and $\hat{\beta}_1$.

- Find the expected value of $\hat{\beta}_1$ under the true model and hence find the bias in estimation of β_1 .
- Under what condition on the points x_{1i}, x_{2i} is $\hat{\beta}_1$ unbiased?

Solution 1

Under the true model, we have the following,

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i, \quad E(\epsilon_i) = 0, \quad \text{Var}(\epsilon_i) = \sigma^2,$$

with x_{1i}, x_{2i} fixed. We regress y on x_1 alone (omitting x_2) by least squares. Writing

$$S_{x_1 x_1} = \sum_{i=1}^n (x_{1i} - \bar{x}_1)^2, \quad S_{x_1 x_2} = \sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2), \quad S_{x_1 y} = \sum_{i=1}^n (x_{1i} - \bar{x}_1)(y_i - \bar{y}),$$

the least squares slope from this simple regression is

$$\hat{\beta}_1 = \frac{S_{x_1 y}}{S_{x_1 x_1}}.$$

Now, we'll calculate the expected value and bias for $\hat{\beta}_1$. Substituting the true model for y_i ,

$$\begin{aligned} S_{x_1 y} &= \sum_{i=1}^n (x_{1i} - \bar{x}_1)(y_i - \bar{y}) \\ &= \sum_{i=1}^n (x_{1i} - \bar{x}_1) [\beta_1 (x_{1i} - \bar{x}_1) + \beta_2 (x_{2i} - \bar{x}_2) + (\epsilon_i - \bar{\epsilon})] \\ &= \beta_1 S_{x_1 x_1} + \beta_2 S_{x_1 x_2} + \sum_{i=1}^n (x_{1i} - \bar{x}_1)(\epsilon_i - \bar{\epsilon}). \end{aligned}$$

Since the x_{1i}, x_{2i} are fixed and $E(\epsilon_i) = 0$ for every i , taking expectations makes the last term vanish:

$$E(S_{x_1 y}) = \beta_1 S_{x_1 x_1} + \beta_2 S_{x_1 x_2}.$$

Hence

$$E(\hat{\beta}_1) = \frac{E(S_{x_1 y})}{S_{x_1 x_1}} = \beta_1 + \beta_2 \frac{S_{x_1 x_2}}{S_{x_1 x_1}}.$$

Note that $S_{x_1 x_2}/S_{x_1 x_1}$ is exactly the least squares slope $b_{x_2 x_1}$ obtained by regressing x_2 on x_1 . Thus

$$E(\hat{\beta}_1) = \beta_1 + \beta_2 \frac{S_{x_1 x_2}}{S_{x_1 x_1}},$$

$$\text{Bias}(\hat{\beta}_1) = E(\hat{\beta}_1) - \beta_1 = \beta_2 \frac{S_{x_1 x_2}}{S_{x_1 x_1}}.$$

(b) Condition for unbiasedness. From the expression obtained above, $\hat{\beta}_1$ is unbiased (i.e. $E(\hat{\beta}_1) = \beta_1$) if and only if

$$\beta_2 S_{x_1 x_2} = 0.$$

Excluding the trivial case $\beta_2 = 0$ (where x_2 has no effect on y to begin with), $\hat{\beta}_1$ is unbiased precisely when

$$S_{x_1 x_2} = \sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2) = 0,$$

i.e. when x_1 and x_2 are *uncorrelated in the sample*. If the fixed values x_{1i} and x_{2i} have zero sample covariance, omitting x_2 from the regression does not bias the estimate of β_1 .

Question 2

Consider two random variables (X, Y) distributed as bivariate normal. Based on a random sample from this bivariate distribution, the fitted least squares regression line of Y on X and that of X on Y were as follows:

$$Y = 22 - 3X \quad X = 5.84 - 0.12Y$$

Find the sample means of X and Y , the sample correlation between X and Y , and the ratio of the sample variances of X and Y .

[Normality is not required here except to indicate that both X and Y are random and continuous.]

Solution 2

We are given the two fitted regression lines:

$$\text{Regression of } Y \text{ on } X : Y = 22 - 3X, \quad b_{YX} = -3,$$

$$\text{Regression of } X \text{ on } Y : X = 5.84 - 0.12Y, \quad b_{XY} = -0.12.$$

- (a) Both regression lines pass through the point of sample means $(\bar{X}, \bar{Y})$, so $(\bar{X}, \bar{Y})$ is precisely the point of intersection of the two lines. Writing both equations in the same form,

$$\begin{aligned} 3X + Y &= 22, \\ X + 0.12Y &= 5.84. \end{aligned}$$

From the first equation, $Y = 22 - 3X$. Substituting into the second equation,

$$X + 0.12(22 - 3X) = 5.84 \implies X + 2.64 - 0.36X = 5.84 \implies 0.64X = 3.2.$$

Hence $X = 5$ and $Y = 22 - 3(5) = 7$. So

$$\bar{X} = 5 \quad \bar{Y} = 7$$

- (b) Also we know that for simple linear regression, the two slopes are related to the correlation coefficient r by

$$b_{YX} = r \frac{\sigma_Y}{\sigma_X}, \quad b_{XY} = r \frac{\sigma_X}{\sigma_Y},$$

so that

$$r^2 = b_{YX} \cdot b_{XY} = (-3)(-0.12) = 0.36 \implies r = \pm 0.6.$$

Since b_{YX} and b_{XY} are both negative, r must also be negative (the correlation coefficient always shares the common sign of the two regression slopes). Hence

$$r = -0.6.$$

(c) To calculate the ratio of sample variance, we start by dividing the two slopes,

$$\frac{b_{YX}}{b_{XY}} = \frac{r \sigma_Y / \sigma_X}{r \sigma_X / \sigma_Y} = \frac{\sigma_Y^2}{\sigma_X^2}.$$

Substituting the given values,

$$\frac{\sigma_Y^2}{\sigma_X^2} = \frac{-3}{-0.12} = 25.$$

So

$$\boxed{\frac{\text{Var}(Y)}{\text{Var}(X)} = 25} \quad \left(\text{equivalently } \frac{\text{Var}(X)}{\text{Var}(Y)} = \frac{1}{25} = 0.04 \right).$$

Question 3

Use the `anscombe` dataset available in R. It consists of 4 sets of data, each with 11 bivariate observations. Compute the following statistics for each dataset: means of the two variables, variances of the two variables, correlation coefficient between the two variables, regression line and R^2 of the second variable on the first. Plot the four datasets and comment on the suitability of a linear regression model in each case.

[This exercise demonstrates the need for plotting before modeling the data.]

Solution 3

The `anscombe` dataset in R contains four pairs of variables $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$, each with 11 observations. For each dataset i , we compute the sample mean and variance of x_i and y_i , the correlation coefficient r_i , and fit the simple linear regression of y_i on x_i .

Calculating the Summary Statistics

Listing 1: Means, variances and correlations

```
1 data <- anscombe
2
3 mean_table <- data.frame(
4   Dataset = c("Dataset 1", "Dataset 2", "Dataset 3", "Dataset 4")
5   ,
6   X = c(mean(data$x1), mean(data$x2), mean(data$x3), mean(data$x4))
7   ,
8   Y = c(mean(data$y1), mean(data$y2), mean(data$y3), mean(data$y4))
9   )
```

```

8
9 var_table <- data.frame(
10   Dataset = c("Dataset 1", "Dataset 2", "Dataset 3", "Dataset 4")
11   ,
12   X = c(var(data$x1), var(data$x2), var(data$x3), var(data$x4)),
13   Y = c(var(data$y1), var(data$y2), var(data$y3), var(data$y4))
14 )
15
16 corr_table <- data.frame(
17   Dataset = c("Dataset 1", "Dataset 2", "Dataset 3", "Dataset 4")
18   ,
19   Corr = c(cor(data$x1, data$y1), cor(data$x2, data$y2),
20           cor(data$x3, data$y3), cor(data$x4, data$y4))
21 )

```

Dataset	Mean X	Mean Y	Var X	Var Y	Corr.
Dataset 1	9	7.5009	11	4.1273	0.8164
Dataset 2	9	7.5009	11	4.1276	0.8162
Dataset 3	9	7.5000	11	4.1226	0.8163
Dataset 4	9	7.5009	11	4.1233	0.8165

Table 1: Sample means, variances and correlation coefficients for the four Anscombe datasets.

Fitting the regression lines

Listing 2: Regression of y on x for each dataset

```

1 mod1 <- lm(y1 ~ x1, data = anscombe)
2 mod2 <- lm(y2 ~ x2, data = anscombe)
3 mod3 <- lm(y3 ~ x3, data = anscombe)
4 mod4 <- lm(y4 ~ x4, data = anscombe)
5
6 model_table <- data.frame(
7   Dataset = c("Dataset 1", "Dataset 2", "Dataset 3", "Dataset 4")
8   ,
9   Intercept = c(coef(mod1)[1], coef(mod2)[1], coef(mod3)[1], coef(
10     mod4)[1]),
11   Slope = c(coef(mod1)[2], coef(mod2)[2], coef(mod3)[2], coef(
12     mod4)[2]),
13   R_Squared = c(summary(mod1)$r.squared, summary(mod2)$r.squared,
14                 summary(mod3)$r.squared, summary(mod4)$r.squared)
15   ,
16   row.names = NULL
17 )

```

Dataset	Intercept	Slope	R^2
Dataset 1	3.0001	0.5001	0.6665
Dataset 2	3.0009	0.5000	0.6662
Dataset 3	3.0025	0.4997	0.6663
Dataset 4	3.0017	0.4999	0.6667

Table 2: Fitted regression line $\hat{y} = b_0 + b_1x$ and R^2 for each dataset.

All four datasets share the same mean of x (9), mean of y (≈ 7.50), variance of x (11), variance of y (≈ 4.13), correlation (≈ 0.816), fitted regression line ($\hat{y} \approx 3 + 0.5x$), and R^2 (≈ 0.667). If one relied on these summary statistics alone, the four datasets would appear statistically identical.

Plotting the datasets

Listing 3: Scatterplot with fitted regression lines

```

1 library(ggplot2)
2
3 anscombe_long <- data.frame(
4   Dataset = rep(c("Dataset 1", "Dataset 2", "Dataset 3", "Dataset
5     4"), each = 11),
6   x = c(anscombe$x1, anscombe$x2, anscombe$x3, anscombe$x4),
7   y = c(anscombe$y1, anscombe$y2, anscombe$y3, anscombe$y4)
8 )
9
10 ggplot(anscombe_long, aes(x = x, y = y)) +
11   geom_point(color = "#2E86C1", size = 3, alpha = 0.8) +
12   geom_smooth(method = "lm", formula = y ~ x, color = "#CB4335",
13     se = FALSE, linewidth = 1.2) +
14   facet_wrap(~ Dataset) +
15   coord_cartesian(xlim = c(4, 19), ylim = c(3, 13)) +
16   scale_x_continuous(breaks = seq(4, 18, 2)) +
17   scale_y_continuous(breaks = seq(4, 12, 2)) +
18   labs(
19     title = "Anscombe's Quartet",
20     x = "X Values",
21     y = "Y Values"
22   ) +
23   theme_minimal(base_size = 14) +
24   theme(
25     strip.background = element_rect(fill = "#EAEDED", color = NA),
26     strip.text = element_text(face = "bold", size = 12),
27     panel.border = element_rect(fill = NA, color = "#D5D8DC"),
28     panel.grid.minor = element_blank(),
29     plot.title = element_text(face = "bold", hjust = 0.5, margin
    = margin(b = 15))
30   )

```

Comment on the suitability of a linear model

- **Dataset 1** shows a roughly linear relationship with points scattered randomly around the fitted line. A simple linear regression model is well suited here.
- **Dataset 2** is clearly *curvilinear* (an inverted-parabola shape). A linear fit is misleading despite giving the same R^2 ; a quadratic or other nonlinear model would be far more appropriate.
- **Dataset 3** has an almost perfect linear relationship except for a single outlier that pulls the fitted line away from the rest of the data. Removing the outlier would give a near-perfect linear fit; a robust regression method would also handle this well.
- **Dataset 4** has all x values identical except one, so the fitted slope is driven entirely by a single high-leverage point. The regression line is not meaningful here, and a linear model should not be trusted.

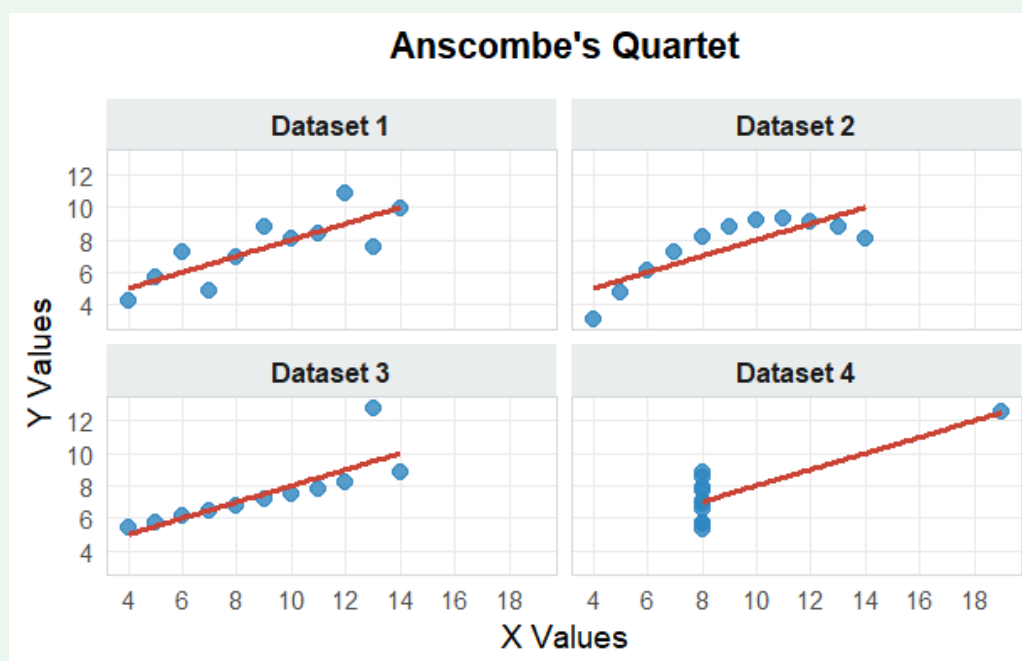


Figure 1: Scatterplots of the four Anscombe datasets with fitted least squares lines.

We see that identical summary statistics (means, variances, correlation, regression coefficients, R^2) can arise from datasets with drastically different underlying structures. Thus we conclude that **numerical summaries alone are not sufficient to validate a linear regression model and hence visualizing the data is essential before fitting any model.**

Question 4

Suppose you have fitted a regression line $Y = b_0 + b_1X$ on n pairs of observations where b_0 and b_1 are the least squares estimators. For a new value x_0 of X , what is the predicted

value of Y ? What is the variance of this prediction? How does the variance depend on x_0 ? Interpret this result.

Solution 4

Predicted value. For a new value x_0 of X , the predicted value of Y is simply the fitted line evaluated at x_0 :

$$\hat{Y}_0 = b_0 + b_1 x_0.$$

Since $b_0 = \bar{y} - b_1 \bar{x}$, this can equivalently be written as

$$\hat{Y}_0 = \bar{y} + b_1(x_0 - \bar{x}).$$

Consider the following

$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$. Since $b_1 = \frac{\sum_i (x_i - \bar{x}) y_i}{S_{xx}}$ (because $\sum_i (x_i - \bar{x}) \bar{y} = 0$) and $\bar{y} = \frac{1}{n} \sum_i y_i$, we can write $\hat{Y}_0$ as a linear combination of the y_i 's:

$$\hat{Y}_0 = \sum_{i=1}^n \left[\frac{1}{n} + \frac{(x_i - \bar{x})(x_0 - \bar{x})}{S_{xx}} \right] y_i = \sum_{i=1}^n c_i y_i, \quad c_i = \frac{1}{n} + \frac{(x_i - \bar{x})(x_0 - \bar{x})}{S_{xx}}.$$

Under the usual assumptions that the y_i are independent with common variance σ^2 ,

$$\text{Var}(\hat{Y}_0) = \sigma^2 \sum_{i=1}^n c_i^2.$$

Expanding c_i^2 and using $\sum_i (x_i - \bar{x}) = 0$ and $\sum_i (x_i - \bar{x})^2 = S_{xx}$,

$$\begin{aligned} \sum_{i=1}^n c_i^2 &= \sum_{i=1}^n \left[\frac{1}{n^2} + \frac{2(x_i - \bar{x})(x_0 - \bar{x})}{n S_{xx}} + \frac{(x_i - \bar{x})^2 (x_0 - \bar{x})^2}{S_{xx}^2} \right] \\ &= \frac{1}{n} + \underbrace{\frac{2(x_0 - \bar{x})}{n S_{xx}} \sum_{i=1}^n (x_i - \bar{x})}_{=0} + \frac{(x_0 - \bar{x})^2}{S_{xx}^2} \underbrace{\sum_{i=1}^n (x_i - \bar{x})^2}_{=S_{xx}} \\ &= \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}. \end{aligned}$$

Hence

$$\text{Var}(\hat{Y}_0) = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right].$$

The variance is a quadratic, upward-opening function of x_0 , minimized at $x_0 = \bar{x}$, where

$$\text{Var}(\hat{Y}_0) \Big|_{x_0=\bar{x}} = \frac{\sigma^2}{n},$$

the smallest possible variance for any predicted point on the line. As x_0 moves away from $\bar{x}$ in either direction, $(x_0 - \bar{x})^2$ increases, so $\text{Var}(\hat{Y}_0)$ increases thus growing without bound as $|x_0 - \bar{x}| \rightarrow \infty$.

Thus we conclude that predictions are most precise near the centre of the observed x -data and become increasingly unreliable as x_0 moves toward the edges of the observed range, and especially outside it. Extrapolating the fitted line to values of x_0 far from $\bar{x}$ should be treated with caution, since the uncertainty in the prediction grows rapidly the farther x_0 is from the bulk of the data used to fit the line.